TCP Connection Performance Analysis using Mininet and Wireshark

Assignment 2

Submitted By

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1. Objective

The objective of this assignment is to analyze TCP communication performance using Mininet and Wireshark. Two TCP communication modes were implemented and compared:

1. Persistent TCP Connection
2. New TCP Connection per Message

Performance was evaluated using response time, throughput measurements, graphs, and packet analysis using Wireshark.

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2. Mininet Topology

The following Mininet topology was used:

h1 ---- s1 ---- h2

Where:

- h1 acts as TCP Server
- h2 acts as TCP Client
- Switch s1 connects both hosts

Network configuration:

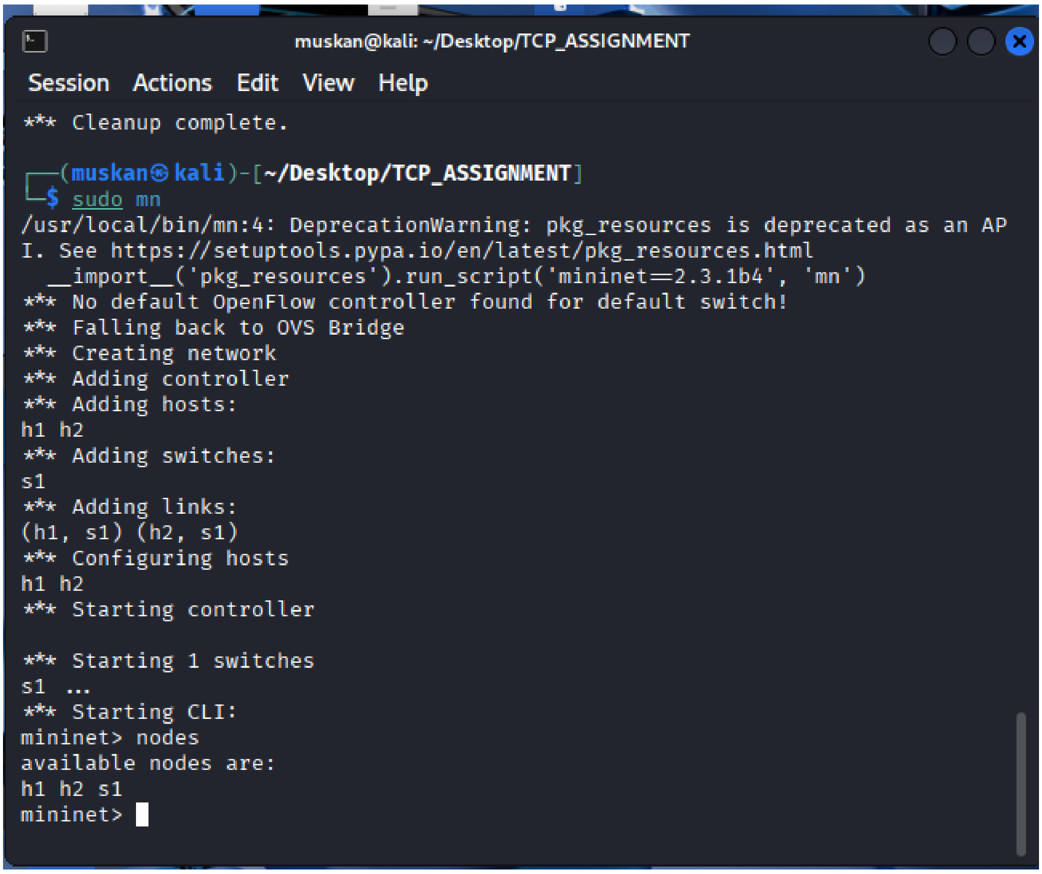
- Bandwidth = 5 Mbps
- Delay = 50 ms

Command used:

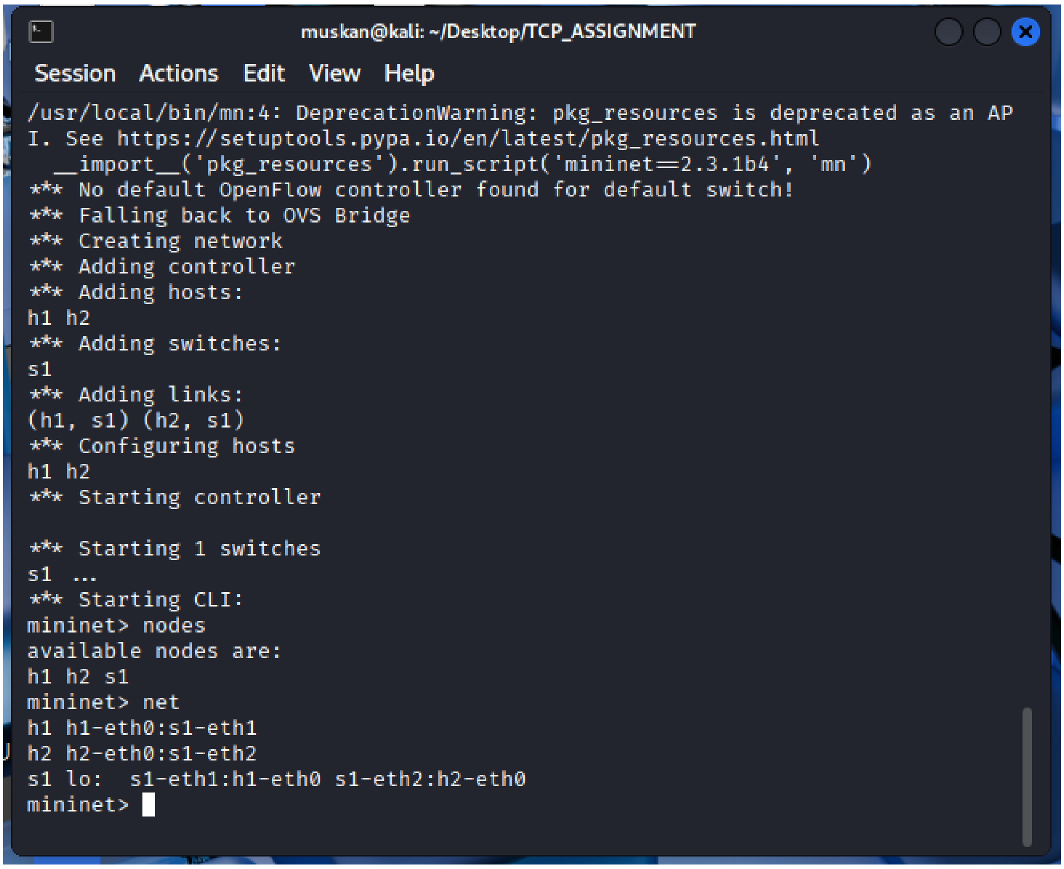
sudo mn --link tc,bw=5,delay=50ms

Insert screenshots:

- nodes.png


muskan@kali: ~/Desktop/TCP_ASSIGNMENT
Session Actions Edit View Help
*** Cleanup complete.
(muskan㉿kali)-[~/Desktop/TCP_ASSIGNMENT]
$ sudo mn
/usr/local/bin/mn:4: DeprecationWarning: pkg_resources is deprecated as an AP
I. See https://setuptools.pypa.io/en/latest/pkg_resources.html
__import__('pkg_resources').run_script('mininet==2.3.1b4', 'mn')
*** No default OpenFlow controller found for default switch!
*** Falling back to OVS Bridge
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1)
*** Configuring hosts
h1 h2
*** Starting controller
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> nodes
available nodes are:
h1 h2 s1
mininet>

-
- net.png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*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1)
*** Configuring hosts
h1 h2
*** Starting controller
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> nodes
available nodes are:
h1 h2 s1
mininet> net
h1 h1-eth0:s1-eth1
h2 h2-eth0:s1-eth2
s1 lo: s1-eth1:h1-eth0 s1-eth2:h2-eth0
mininet>

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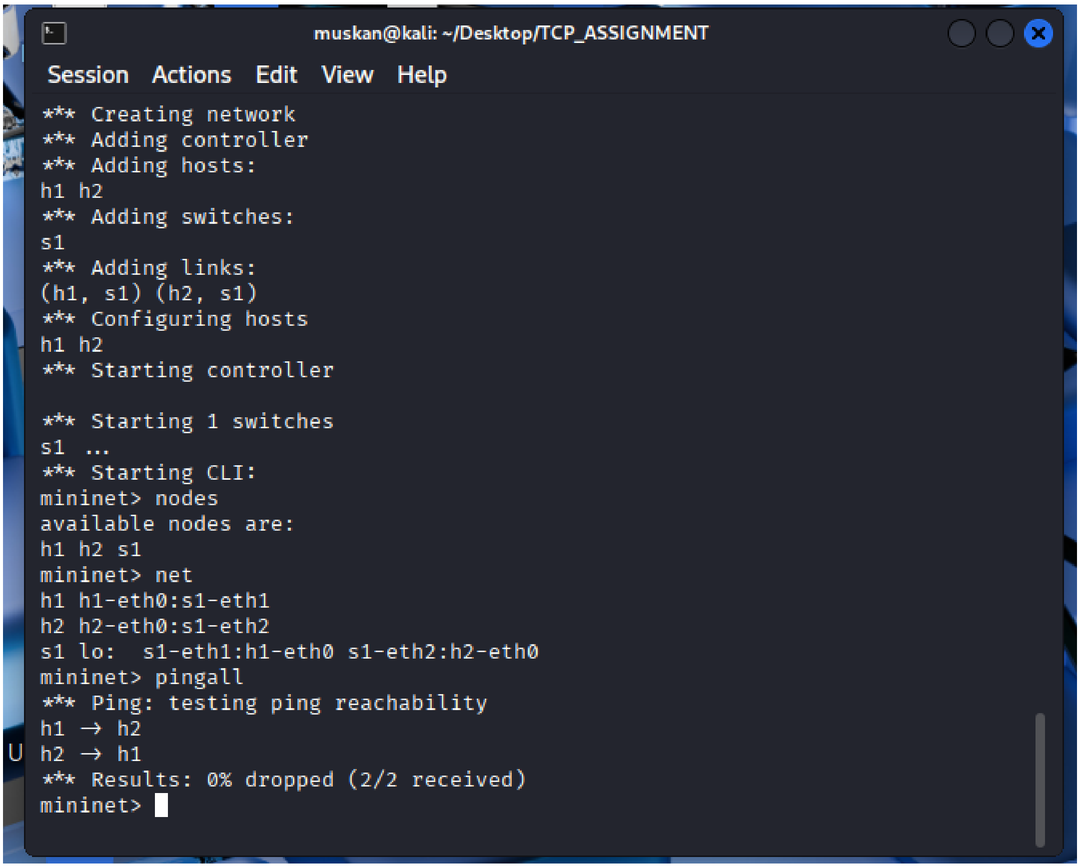
- pingall.png



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3. TCP Client Server Communication

The server listens on TCP port 5000 and accepts incoming client connections.

Message format:

MSG_ID|MESSAGE_SIZE|MESSAGE_DATA

Example:

3|512|AAAAAAA...

The server processes each message and responds with:

ACK|MSG_ID|RECEIVED_SIZE

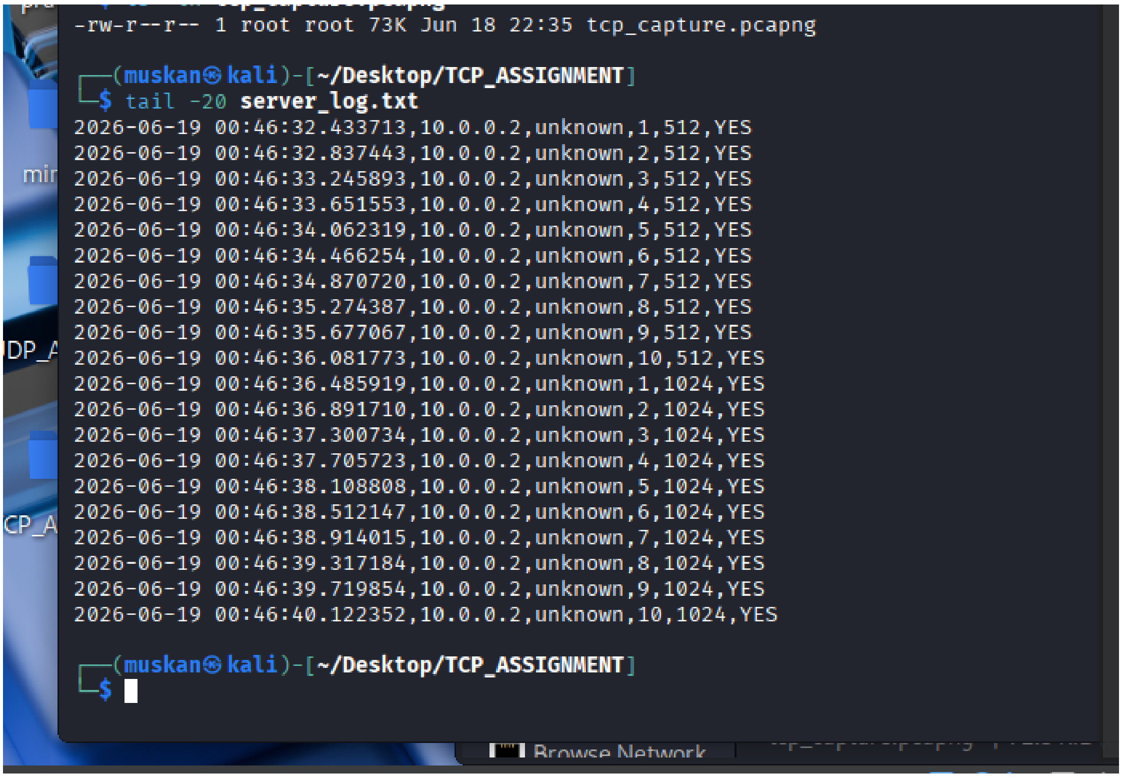
Example:

ACK|3|512

The client measures the response time for every message and calculates throughput.

Insert screenshots:

- server_output.png

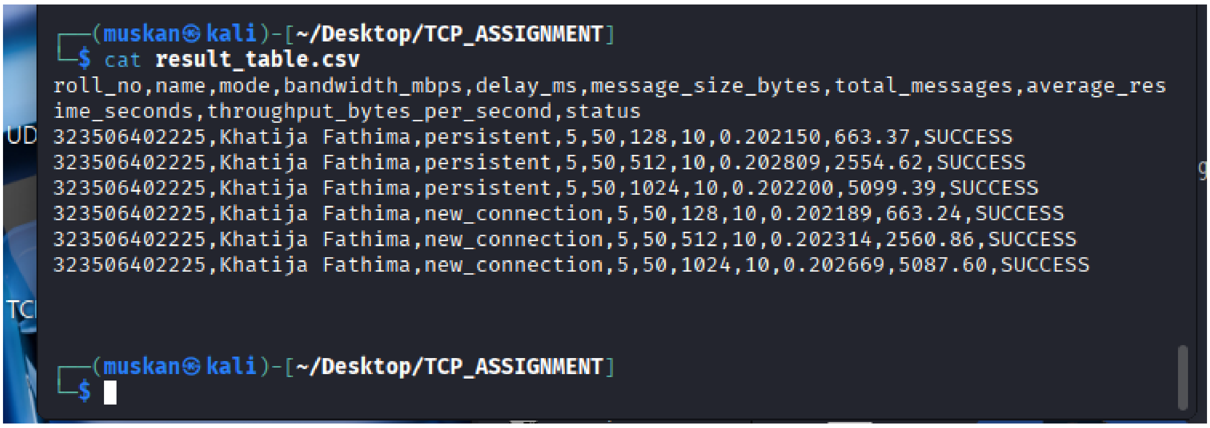


```
-rw-r--r-- 1 root root 73K Jun 18 22:35 tcp_capture.pcapng

┌──(muskan㉿kali)-[~/Desktop/TCP_ASSIGNMENT]
└─$ tail -20 server_log.txt
2026-06-19 00:46:32.433713,10.0.0.2,unknown,1,512,YES
2026-06-19 00:46:32.837443,10.0.0.2,unknown,2,512,YES
2026-06-19 00:46:33.245893,10.0.0.2,unknown,3,512,YES
2026-06-19 00:46:33.651553,10.0.0.2,unknown,4,512,YES
2026-06-19 00:46:34.062319,10.0.0.2,unknown,5,512,YES
2026-06-19 00:46:34.466254,10.0.0.2,unknown,6,512,YES
2026-06-19 00:46:34.870720,10.0.0.2,unknown,7,512,YES
2026-06-19 00:46:35.274387,10.0.0.2,unknown,8,512,YES
2026-06-19 00:46:35.677067,10.0.0.2,unknown,9,512,YES
2026-06-19 00:46:36.081773,10.0.0.2,unknown,10,512,YES
2026-06-19 00:46:36.485919,10.0.0.2,unknown,1,1024,YES
2026-06-19 00:46:36.891710,10.0.0.2,unknown,2,1024,YES
2026-06-19 00:46:37.300734,10.0.0.2,unknown,3,1024,YES
2026-06-19 00:46:37.705723,10.0.0.2,unknown,4,1024,YES
2026-06-19 00:46:38.108808,10.0.0.2,unknown,5,1024,YES
2026-06-19 00:46:38.512147,10.0.0.2,unknown,6,1024,YES
2026-06-19 00:46:38.914015,10.0.0.2,unknown,7,1024,YES
2026-06-19 00:46:39.317184,10.0.0.2,unknown,8,1024,YES
2026-06-19 00:46:39.719854,10.0.0.2,unknown,9,1024,YES
2026-06-19 00:46:40.122352,10.0.0.2,unknown,10,1024,YES

┌──(muskan㉿kali)-[~/Desktop/TCP_ASSIGNMENT]
└─$
```

- client_output.png



```
┌──(muskan㉿kali)-[~/Desktop/TCP_ASSIGNMENT]
└─$ cat result_table.csv
roll_no,name,mode,bandwidth_mbps,delay_ms,message_size_bytes,total_messages,average_response_time_seconds,throughput_bytes_per_second,status
323506402225,Khatija Fathima,persistent,5,50,128,10,0.202150,663.37,SUCCESS
323506402225,Khatija Fathima,persistent,5,50,512,10,0.202809,2554.62,SUCCESS
323506402225,Khatija Fathima,persistent,5,50,1024,10,0.202200,5099.39,SUCCESS
323506402225,Khatija Fathima,new_connection,5,50,128,10,0.202189,663.24,SUCCESS
323506402225,Khatija Fathima,new_connection,5,50,512,10,0.202314,2560.86,SUCCESS
323506402225,Khatija Fathima,new_connection,5,50,1024,10,0.202669,5087.60,SUCCESS

┌──(muskan㉿kali)-[~/Desktop/TCP_ASSIGNMENT]
└─$
```

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4. Persistent vs New Connection Mode

Persistent Connection

- One TCP connection is established.
- Multiple messages are sent using the same connection.
- Connection is closed after all messages are transmitted.

Advantages:

- Lower connection overhead
- Fewer TCP handshakes

- Better efficiency

New Connection Mode

- A new TCP connection is created for every message.
- Each message requires a separate TCP handshake and connection termination.

Advantages:

- Independent transactions
- Better fault isolation

Disadvantages:

- Higher overhead
- More TCP handshakes
- Increased resource usage

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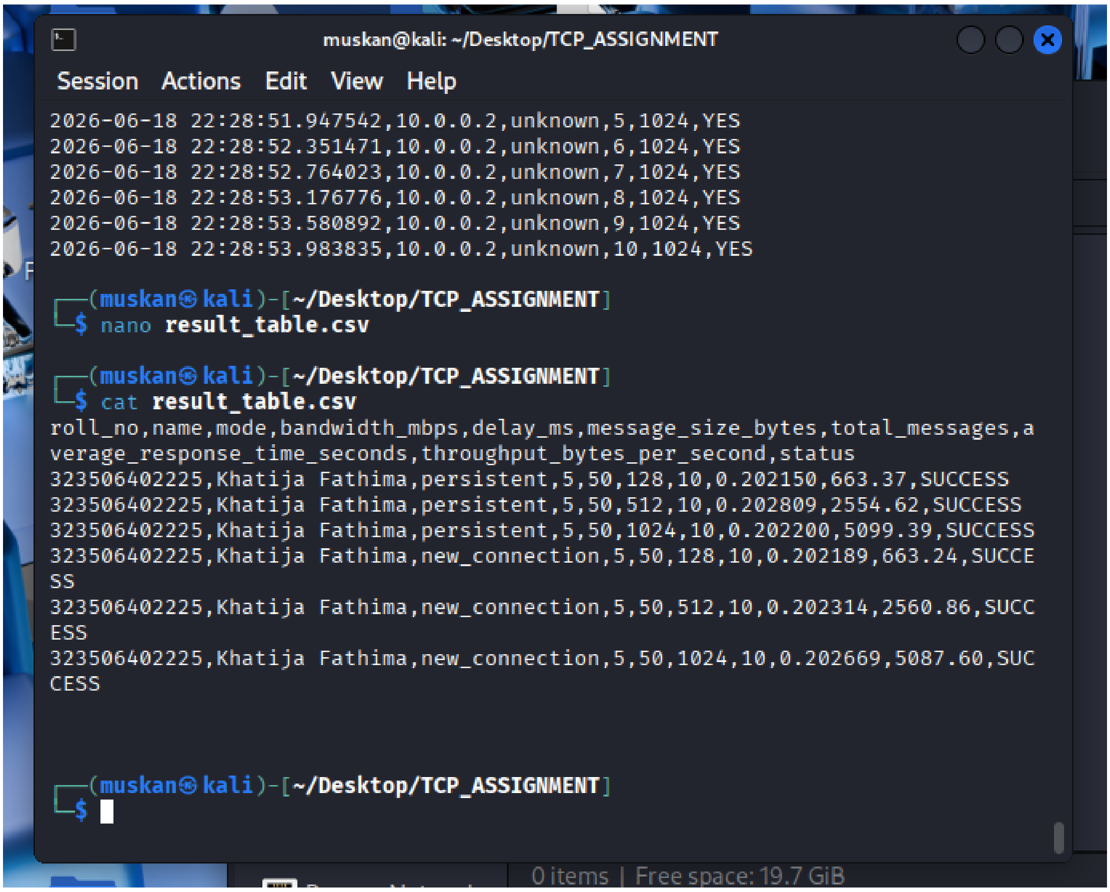
5. Experimental Results

Insert result_table.csv



Observation:

The average response time for both modes is approximately 0.20 seconds.

Throughput increases significantly as message size increases from 128 bytes to 1024 bytes.

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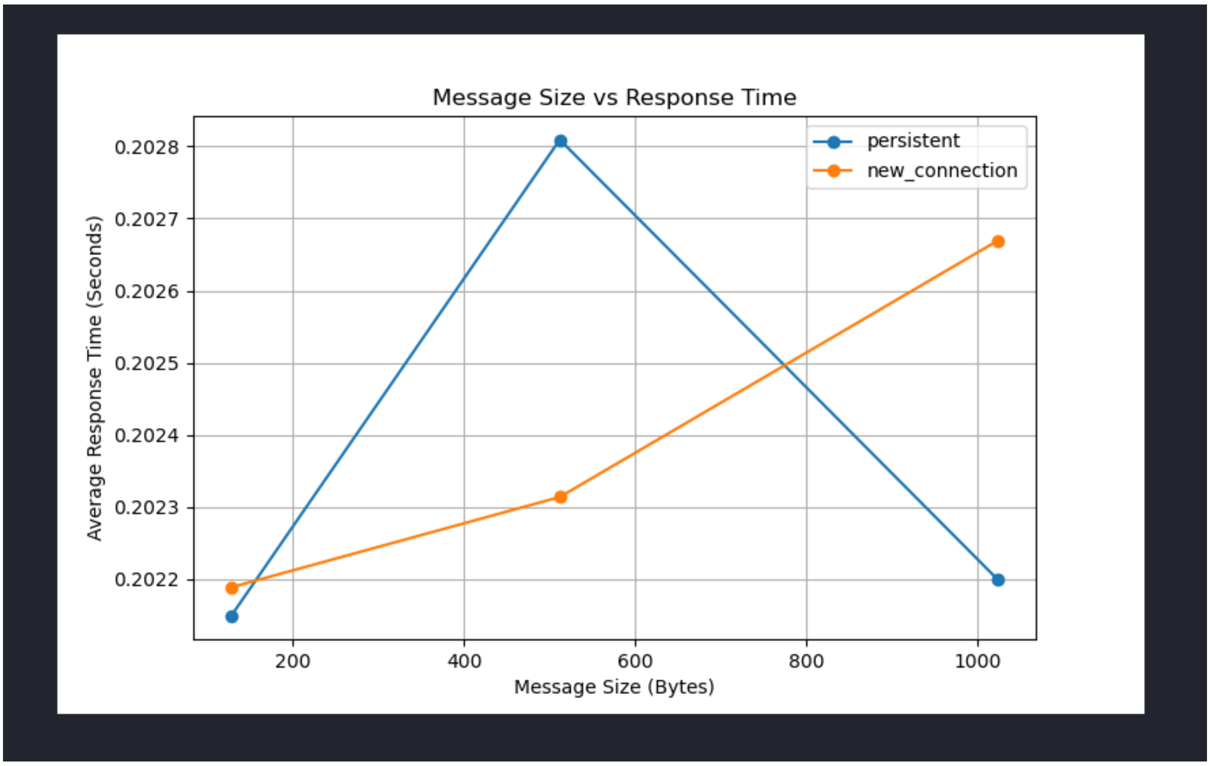
6. Graph Analysis

Insert:

Graph 1

mode_vs_response_time.png

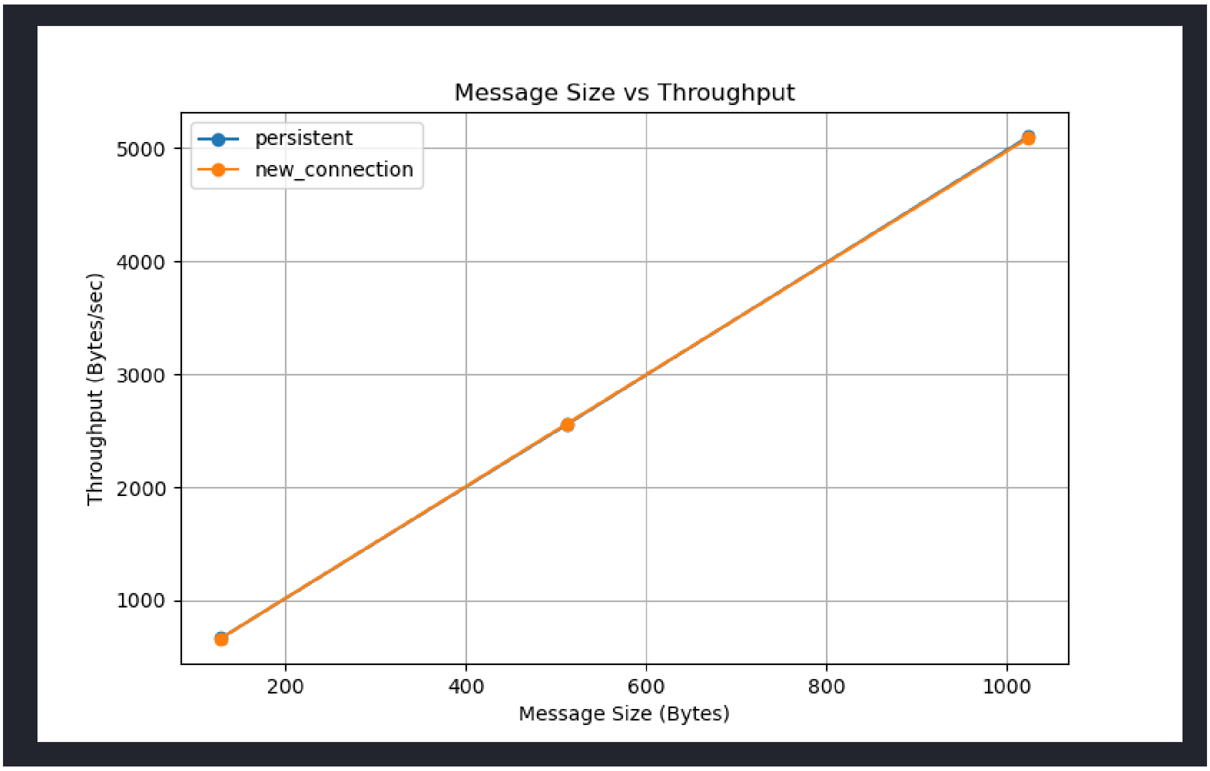


Observation:

Persistent mode shows slightly better response time because connection setup occurs only once.

Graph 2

message_size_vs_throughput.png



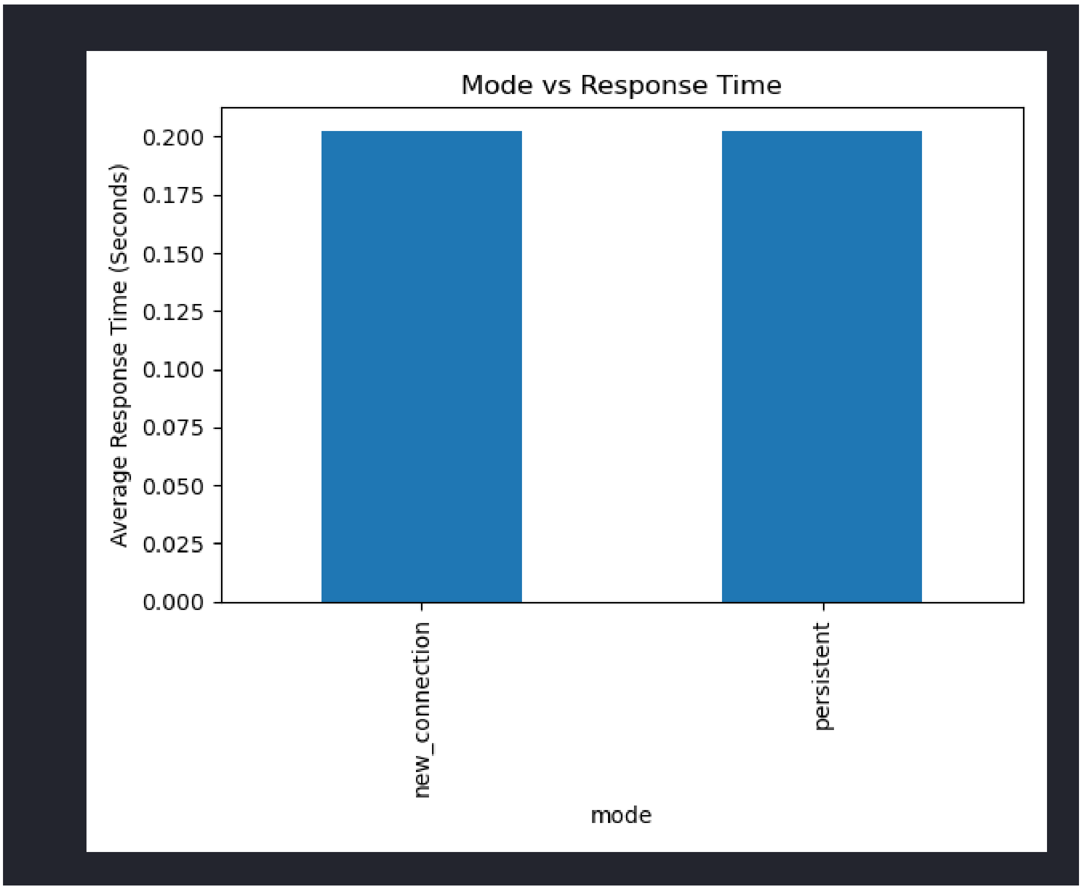
Observation:

Throughput increases as message size increases.

1024-byte messages achieve the highest throughput.

Graph 3

message_response_time.png



Observation:

Response times remain relatively stable for all messages with minor fluctuations.

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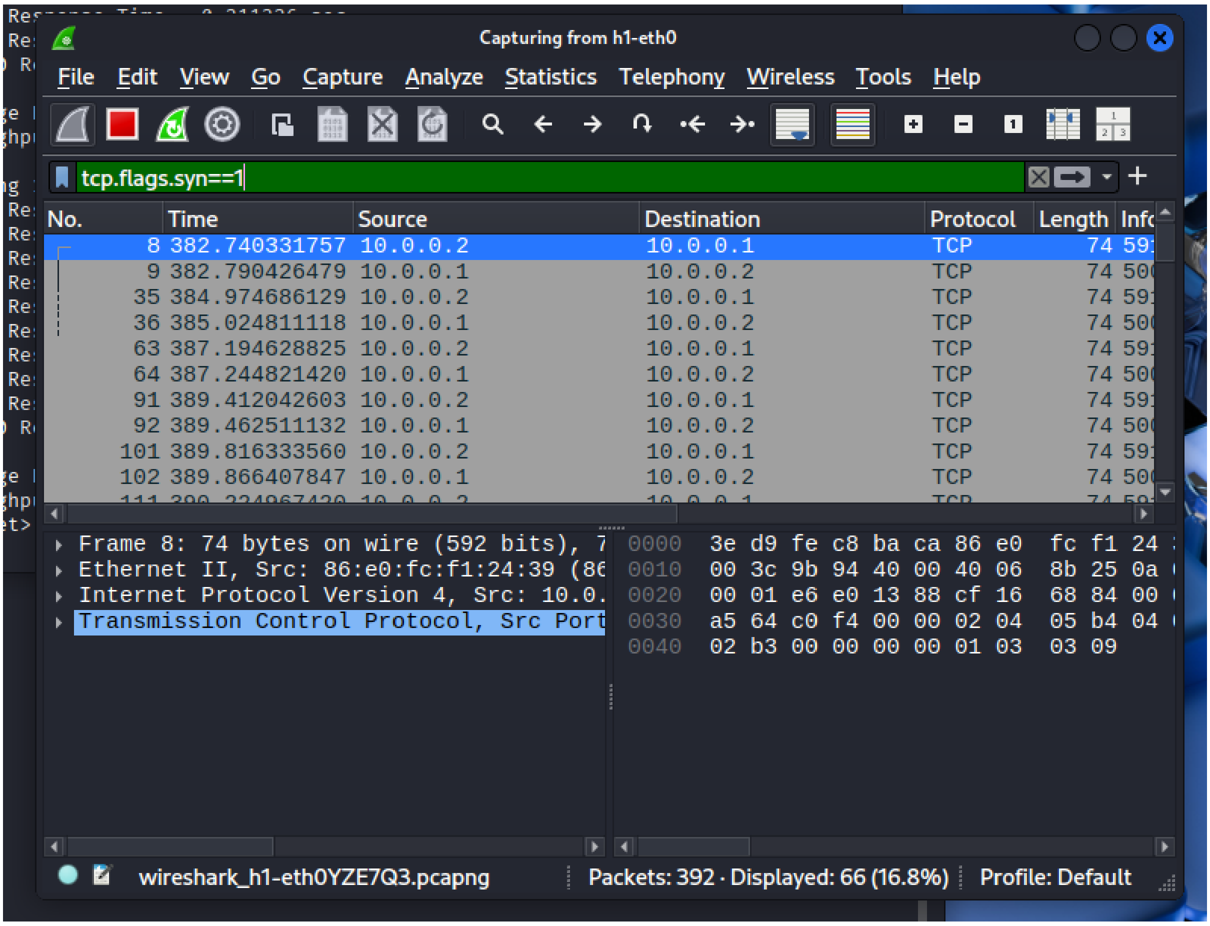
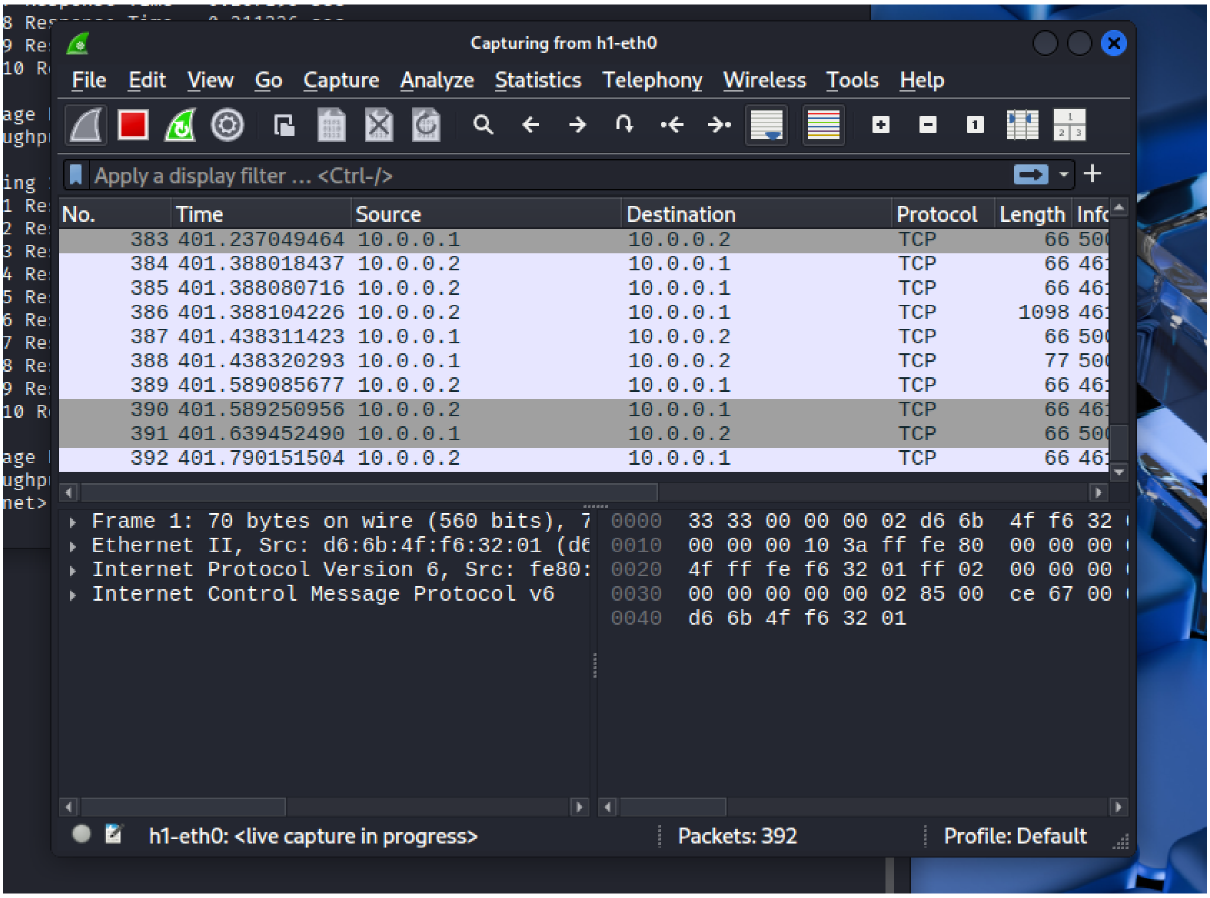
7. Wireshark Analysis

Insert screenshots:

- persistent_handshake.png



- persistent_data_packets.png



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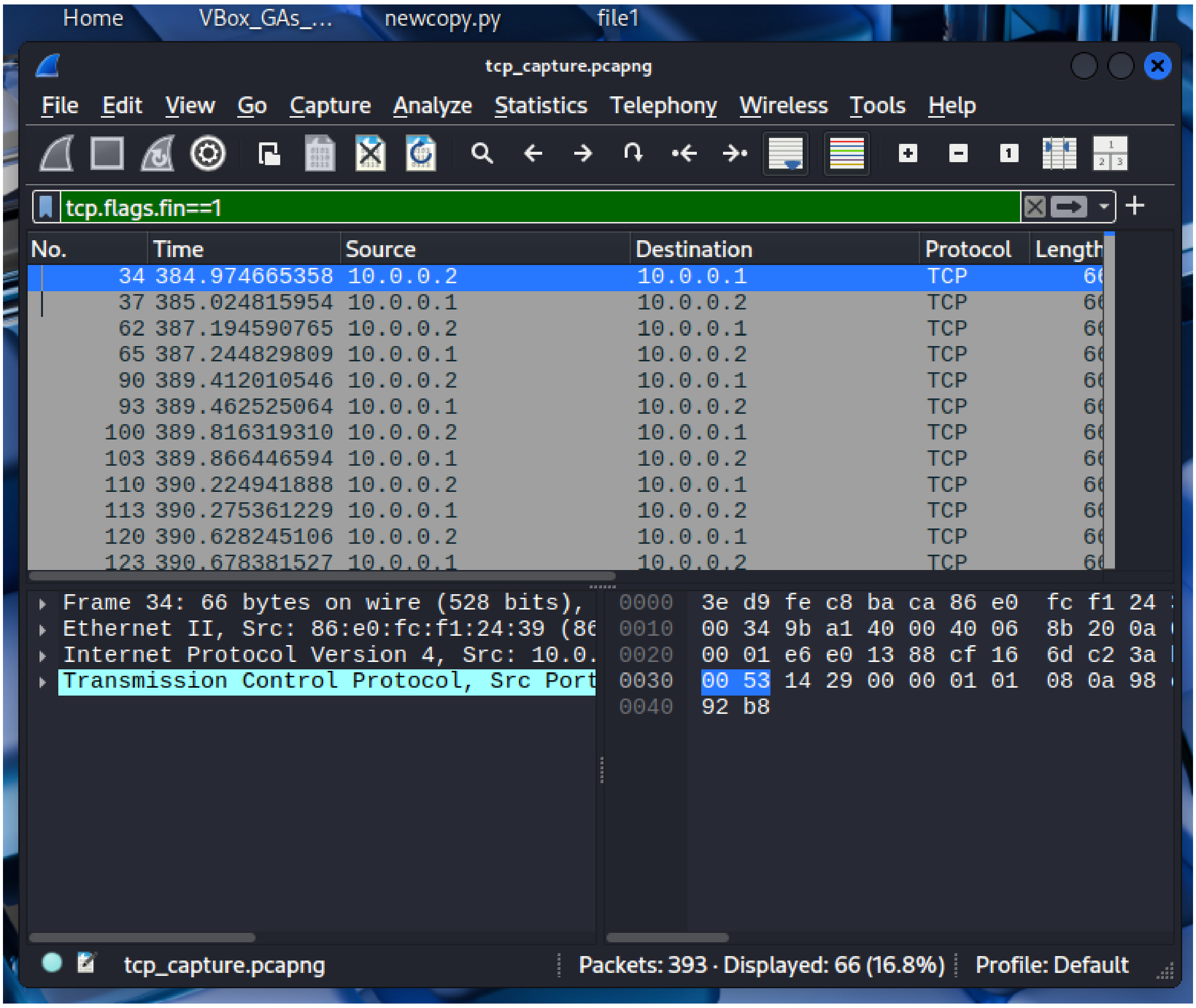
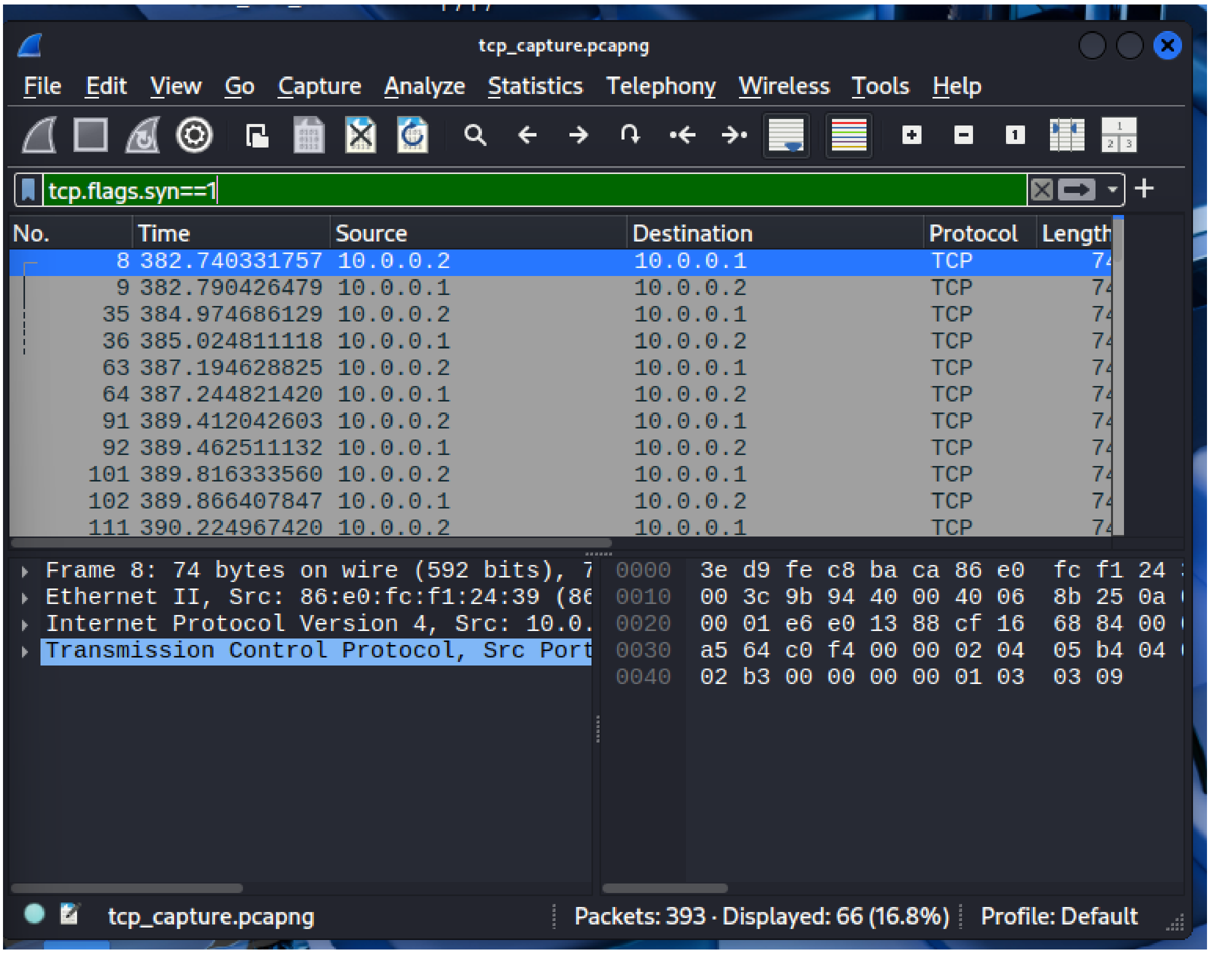
- persistent_connection_close.png

- 

Home
VBox_GAs_...
newcopy.py
file1
tcp_capture.pcapng
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.flags.fin==1
No. Time Source Destination Protocol Length
34 384.974665358 10.0.0.2 10.0.0.1 TCP
37 385.024815954 10.0.0.1 10.0.0.2 TCP
62 387.194590765 10.0.0.2 10.0.0.1 TCP
65 387.244829809 10.0.0.1 10.0.0.2 TCP
90 389.412010546 10.0.0.2 10.0.0.1 TCP
93 389.462525064 10.0.0.1 10.0.0.2 TCP
100 389.816319310 10.0.0.2 10.0.0.1 TCP
103 389.866446594 10.0.0.1 10.0.0.2 TCP
110 390.224941888 10.0.0.2 10.0.0.1 TCP
113 390.275361229 10.0.0.1 10.0.0.2 TCP
120 390.628245106 10.0.0.2 10.0.0.1 TCP
Frame 34: 66 bytes on wire (528 bits),
Ethernet II, Src: 86:e0:fc:f1:24:39
Internet Protocol Version 4, Src: 10.0.
Transmission Control Protocol, Src Port
0000 3e d9 fe c8 ba ca 86 e0 fc f1 24
0010 00 34 9b a1 40 00 40 06 8b 20 0a
0020 00 01 e6 e0 13 88 cf 16 6d c2 3a
0030 00 53 14 29 00 00 01 01 08 0a 98
0040 92 b8
tcp_capture.pcapng
Packets: 393 · Displayed: 66 (16.8%)
Profile: Default

- new_connection_multiple_handshakes.png



Observations:

Persistent Mode

- One TCP three-way handshake observed.
- Multiple data packets exchanged.
- Single connection termination observed.

New Connection Mode

- Multiple TCP handshakes observed.
- Each message creates a new connection.
- Multiple connection terminations observed.

Wireshark successfully verified the TCP behavior required for the experiment.

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8. Answers to Questions

Why does TCP use a three-way handshake?

TCP uses a three-way handshake to establish a reliable connection between sender and receiver and synchronize sequence numbers before data transmission.

Which mode had lower response time?

Persistent mode had slightly lower response time because connection setup occurs only once.

Why does new-connection mode have more overhead?

Each message requires a new TCP handshake and connection termination, increasing processing time and network overhead.

Which mode is better for CampusChat and why?

Persistent mode is better for CampusChat because users continuously exchange messages and maintaining a connection reduces overhead and improves efficiency.

Which mode is suitable for NetAttend and why?

New-connection mode is suitable for NetAttend because attendance requests are independent transactions and do not require a continuously open connection.

How does message size affect throughput?

As message size increases, throughput increases because more useful data is transmitted per communication cycle.

How did Wireshark help verify TCP behavior?

Wireshark allowed visualization of TCP handshakes, acknowledgments, data transfer, and connection termination packets. It confirmed the correct implementation of both communication modes.

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9. Conclusion

This experiment successfully implemented TCP client-server communication inside Mininet and compared persistent and new-connection modes.

The results show that persistent connections reduce connection overhead and generally provide better efficiency, while new connections introduce additional TCP setup and termination costs. Wireshark successfully verified the TCP behavior observed during the experiment.